A kissing configuration of 12 277 points in dimension 19

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Abstract

We exhibit 12 277 points on the unit sphere $S^{18} \subset \mathbb{R}^{19}$ with pairwise angular distances at least 60°. Hence the kissing number in dimension 19 satisfies $\tau_{19} \geq 12\,277$, one more than the bound 12 276 of K. Li (October 2026). The configuration keeps Li's 10 484-point base, which comes from the odd-sign configuration of Cohn and Li. It restores one more of the 184 base points that Li deletes, and it replaces Li's 1792 added points by 1792 re-optimised points. These form 112 orbits under the order-16 group of sign changes that fixes the restored point. All coordinates are integers. Every pair is checked in exact arithmetic by two independent programs. The statement is also proved in Lean 4 by kernel evaluation without `native_decide`: the inner products among base points are bounded by a formalised structural lemma, and all other inner products are reduced by the symmetry group to finitely many integer inequalities. We also report numerical evidence that restoring a whole further orbit of eight deleted points, as asked by Li, is impossible in this family: the best minimax cosine we reach is $\frac{1}{2} + 2.0 \cdot 10^{-5}$.

1 Introduction

The kissing number τ_n is the largest number of non-overlapping unit balls in $\mathbb{R}^n$ that touch a fixed unit ball. Equivalently, it is the largest size of a set of unit vectors with pairwise inner products at most $1/2$. It is known only for $n \in \{1, 2, 3, 4, 8, 24\}$; see Cohn's table [1] for current bounds and references. For $17 \leq n \leq 23$ the classical lower bounds come from the Leech lattice cross sections Λ_n [9], which give $\tau_{19} \geq 10\,668$. Cohn and Li [2] changed the parity of the sign patterns on the octads and added sign vectors of a code; this gave $\tau_{19} \geq 11\,692$. Ho [3] found a larger code and reached 11 948. In October 2026 K. Li [4, 5] obtained $\tau_{19} \geq 12\,276$. That construction cuts the 20-dimensional Cohn–Li configuration by a hyperplane, deletes 184 base points and adds 1792 reweighted sign vectors with tails. Li lists as open whether "a second orbit of eight deleted points [can] be kept, under Γ or under a smaller group with a larger table" [5, §6]. On October 15, 2026, Cohn's table [1] lists 12 276 as the best lower bound and 24 417 as the best upper bound [10].

Theorem 1.1. *There are* 12 277 *unit vectors in* $\mathbb{R}^{19}$ *with pairwise inner products at most* $1/2$. *Hence* $\tau_{19} \geq 12\,277$.

The configuration of Theorem 1.1 answers a one-point version of Li's question positively: one more deleted point can be kept if the symmetry group is shrunk to its stabiliser and the added points are re-optimised. For a whole extra orbit of eight points we found only configurations with largest cosine $\frac{1}{2} + 2.0 \cdot 10^{-5}$ (Section 5). To our knowledge, as of October 15, 2026, Theorem 1.1 is

the best lower bound for τ_{19}. We checked Cohn's table, the four public repositories that worked on dimensions 17–31 in September and October 2026 [4, 5, 6, 7], the preprint [8] and a web search.

2 Notation and the base

We follow [5]. Coordinates of $\mathbb{R}^{20}$ are numbered $1, \dots, 20$. Let $E \subseteq \mathbb{F}_2^{20}$ be the $[20, 12, 4]$ code spanned by the twelve words of [5, §3.1], which is the Golay code punctured at four coordinates. Its dual contains 130 *octads* (words of weight 8), and two octads meet in at most 4 coordinates. The *odd configuration* B_{20} of Cohn and Li [2] consists of the 760 roots $\pm 2e_i \pm 2e_j$ together with, for every octad o, the 128 vectors with entries ± 1 on o, zeros elsewhere and an odd number of minus signs. All these vectors have squared norm 8.

Lemma 2.1 (Cohn–Li)**.** *Distinct $b, b' \in B_{20}$ satisfy $\langle b, b' \rangle \leq 4$.*

Proof. The two cases that need an argument are the following. Two vectors on the same octad have odd sign patterns, so they differ in an even and positive number of places. Two vectors on different octads overlap in at most four coordinates. The remaining cases are immediate. □

Let $T = \{1, 4, 7, 9\}$, and let $H = \{1, \dots, 20\} \setminus T$ (the *head*). The points of B_{20} orthogonal to $\mathbf{1}_T$ number 10 668. Among them are 192 points on the four octads containing T whose sign pattern on T has two plus and two minus signs. Li deletes 184 of these and keeps the eight points P on the octad $T \cup \{3, 6, 8, 19\}$ with pattern $(+, +, -, -)$ on $(1, 4, 7, 9)$. The map $v \mapsto (v_H, Rv_T/2)$, with

$$R = \begin{pmatrix} 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix},$$

is an isometry from $\mathbf{1}_T^{\perp}$ onto $\mathbb{R}^{19}$ [5, (2)]. We work in $\mathbf{1}_T^{\perp} \subset \mathbb{R}^{20}$ throughout, so "a kissing configuration in $\mathbf{1}_T^{\perp}$" means one in $\mathbb{R}^{19}$.

Li's sign group Γ consists of the 128 sign changes S_g on the coordinates of words g of a 7-dimensional subcode of E missing T [5, §3.2]. Every S_g maps the base onto itself. Under Γ the 184 deleted points form 23 orbits of size 8. In Li's configuration each such orbit conflicts with exactly one orbit of added points, at cosines between 0.5286 and 0.5525.

3 The configuration

Let

$$q_0 = (-1, -1, +1, +1 \text{ on } 1, 4, 7, 9;\ -1, -1, -1, +1 \text{ on } 10, 15, 17, 18;\ 0 \text{ elsewhere}).$$

This is one of Li's deleted points: it lies on the octad $T \cup \{10, 15, 17, 18\}$, has five minus signs and is orthogonal to $\mathbf{1}_T$. Let $\mathcal{H} \leq \Gamma$ be its stabiliser. It has order 16 and consists of the S_g with $g \cap \{10, 15, 17, 18\} = \emptyset$. The configuration $\mathcal{C}$ has three parts.

- **Base:** the 10 668 points of $B_{20} \cap \mathbf{1}_T^{\perp}$ minus the 192 points described above, plus P and q_0. This gives $10\,485 = 492 + 9\,993$ points (492 roots and 9 993 octad vectors). It is $\mathcal{H}$-invariant.
- **New points:** the $\mathcal{H}$-orbits $\{S_g r : S_g \in \mathcal{H}\}$ of 112 integer vectors $r_1, \dots, r_{112} \in \mathbf{1}_T^{\perp} \cap \mathbb{Z}^{20}$. Their entries have absolute value at most $1.16 \cdot 10^8$, and they are listed in the data file `reps112_R20.txt`. Every r_i has all 16 head coordinates nonzero, so each orbit has exactly

16 points, and the 1792 new points have 1792 distinct head sign patterns. These patterns are the head patterns of 1792 words $c \in E$ with $|c \cap T| \in \{1, 2, 3\}$, chosen as one bipartition class in each of the 112 cubes of such words. (The cubes are the cosets of the span of the five tetrads.)

Thus $|\mathcal{C}| = 10\,485 + 1\,792 = 12\,277$.

Proposition 3.1. *All* $12\,277$ *vectors of* $\mathcal{C}$ *are distinct, nonzero and orthogonal to* $\mathbf{1}_T$*. Every pair* $x \neq y$ *satisfies* $\langle x, y\rangle \leq 0$ *or* $4\langle x, y\rangle^2 \leq |x|^2|y|^2$*. Moreover, for every pair involving a new point,* $\frac{1}{4} - \cos^2 \angle(x, y) \geq 4.37 \cdot 10^{-6}$ *whenever the inner product is positive.*

Theorem 1.1 follows by normalising and applying the isometry of Section 2. For the proof of Proposition 3.1 we use the symmetry reduction of [5, §3.3]. Base pairs are covered by Lemma 2.1. A pair $(b, S_g r_i)$ is carried by the isometry S_g to $(S_g b, r_i)$, and $S_g b$ lies in the base. A pair $(S_g r_i, S_h r_j)$ is carried to $(r_i, S_{g+h} r_j)$. It therefore suffices to check two lists:

- the $112 \times 10\,485$ products $\langle r_i, b\rangle$ with b in the base;
- the $112 \cdot 112 \cdot 16$ products $\langle r_i, S_g r_j\rangle$ with $(g, j) \neq (0, i)$.

We verified the whole statement in three independent ways (Section 4).

How the configuration was found. We parametrised Li's family by representatives modulo a sign group and minimised the largest cosine by sequential linear programming on the symmetry-reduced list of pairs. Keeping the base, P and the full Γ-orbit of q_0, and optimising over all 2^7 Γ-invariant choices of kept words, the best minimax value is $\frac{1}{2} + 2.0 \cdot 10^{-5}$. From that configuration we dropped seven of the eight points of the orbit. We then ran a gradient method (Adam on a squared-hinge penalty) on all 1792 added points without symmetry, which reached $\frac{1}{2} + 1.2 \cdot 10^{-5}$. We averaged the result over $\mathcal{H}$ and polished it by $\mathcal{H}$-symmetric sequential linear programming, reaching $\frac{1}{2} - 4.37 \cdot 10^{-6}$. Finally we rounded at scale 10^8. The search played no part in the proof.

4 Verification

Two independent exact programs. The first program computes all $75\,356\,226$ cosines in floating point and re-checks the $2\,390\,370$ pairs with cosine above $\frac{1}{2} - 10^{-6}$ in exact integer arithmetic. The second computes exact 64-bit Gram blocks, which is safe because $|x_k| < 2^{28}$, and checks $4\langle x, y\rangle^2 \leq |x|^2|y|^2$ with unbounded integers for all $26\,085\,106$ pairs with positive inner product. Both pass. Exactly $2\,390\,370$ pairs touch, all of them base–base. Both programs also pass on the 19-coordinate version of the data.

Lean. The certificate uses core Lean 4.35, with no Mathlib needed. Kernel caches accumulate across declarations, so to bound memory the certificate is split into 33 modules, compiled with `lean -o` in dependency order (`lean/mods2/`). They are: generic lemmas; the data; 14 modules of cheap checks; and 16 modules with 14 kernel checks each, taking 6–10 minutes each. The last module contains the main theorem. The same content concatenated into one file is `lean/Main.lean`. The main theorem is

```
theorem K19.kissing19 :
  exists L : List (List Int), L.length = 12277 /\ L.Nodup /\
    (forall v in L, v.length = 20 /\ tsumT v = 0 /\ 0 < dot v v) /\
    L.Pairwise Compat
```

(written with ASCII words for $\exists$, $\wedge$, $\forall$, $\in$). Here `dot` is the integer inner product of lists and `tsumT v` is $v_0 + v_3 + v_6 + v_8$ (the 0-based indices of T). `Compat x y` is $\langle x, y\rangle \le 0 \vee 4\langle x, y\rangle^2 \le \langle x, x\rangle\langle y, y\rangle$. The list L is computed inside Lean from the data: 128 octad records (octad mask and list of sign masks), 492 roots, 16 sign masks and 112 representatives. The proof uses no `sorry`, `axiom`, `admit` or `native_decide`. It has four parts.

1. *Base pairs: a formalised structural proof.* The bit-level identity $|o\cap(m\oplus m')|+2|o\cap m\cap m'| = |o\cap m| + |o\cap m'|$ shows that two odd octad vectors on one octad differ in an even, positive number of coordinates, so their inner product is at most 4. For different octads the inner product is at most $|o\cap o'|$, and the 128^2 octad intersections are checked by `decide`. Roots are handled by a two-term formula.

2. *Group closure.* $\mathcal{H}$ is closed under composition, and every octad block and the root list are closed under the sign changes. Both are checked by `decide`.

3. *Reduced inequalities.* 224 theorems, two per representative, are checked by `decide +kernel`. They cover the 10 485 base products, evaluated through a proved 8-term formula for octad vectors, and the 1 792 products with sign-changed representatives.

4. *Assembly.* The isometry lemmas $\langle S_g x, S_g y\rangle = \langle x, y\rangle$ and $S_g S_h = S_{g\oplus h}$ give `L.Pairwise Compat`. Distinctness follows because no nonzero vector is compatible with itself.

The command `#print axioms K19.kissing19` reports `[propext, Classical.choice, Quot.sound]`. Resource figures are in `lean/lean_output.txt`.

5 Limits and open questions

- *A whole extra orbit.* We ran the minimax optimisation for every combination of three choices: one of the 2^7 Γ-invariant choices of kept words; one of seven candidate deleted orbits; and Li's group Γ or one of the other 15 admissible index-2 subgroups of the sign group of E fixing T (sampled). The best value is $\frac{1}{2} + 2.0\cdot 10^{-5}$. The minima take a handful of discrete values. Passing to subgroups of index 2, 4 or 8 at the optimum does not lower it, and neither does letting P, the extra orbit or base orbits move. This is numerical evidence, not a proof, that Li's question has a negative answer for full orbits.

- *Two extra points.* Two points of the same orbit (same stabiliser $\mathcal{H}$) reach $\frac{1}{2} + 1.75\cdot 10^{-5}$ by $\mathcal{H}$-symmetric optimisation. Whether $\tau_{19} \ge 12\,278$ can be reached this way is open.

- *Dimension 21.* Li's line-weighted points satisfy a rule (his Lemma 4.3) implying that the excluded words in each coset of the line code form a dominating set of the Cayley graph of $\mathbb{F}_2^{10}$ with respect to the 21 lines of $\mathrm{PG}(2,4)$. Hence this family gives at most $4(1024-\gamma)$ points, where γ is the domination number. Tabu search finds dominating sets of size 64 immediately and none of size 63. This suggests $\gamma = 64$, in which case Li's 3840 points would be optimal for line weights. We have no proof.

Use of AI. This work, including the search, the verification code, the Lean formalisation and this text, was carried out by Claude (Anthropic).

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